Information Collection Strategies Experiment

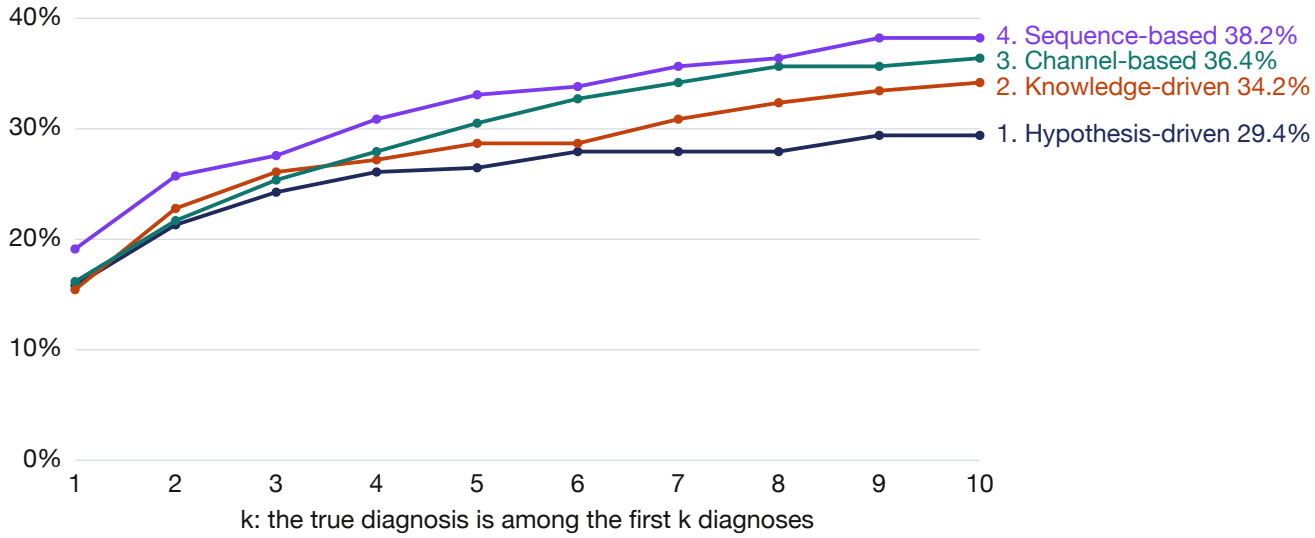
Diagnostic accuracy of four information collection strategies on 272 simulated cases

Run 2026-10-12T16:54:02+00:00 to 2026-10-12T18:03:31+00:00 · record `strategies_272_20261012T165402Z_merged.json` · status completed · commit 35c54b4

Result

Strategy	Hit rate@1	Hit rate@3	Hit rate@5	Hit rate@10	nDCG@10	MRR	Cases
1. Hypothesis-driven	15.8%	24.3%	26.5%	29.4%	0.227	0.205	272
2. Knowledge-driven	15.4%	26.1%	28.7%	34.2%	0.245	0.215	272
3. Channel-based	16.2%	25.4%	30.5%	36.4%	0.256	0.222	272
4. Sequence-based	**19.1%**	**27.6%**	**33.1%**	**38.2%**	**0.282**	**0.250**	272

Bold marks the best value in a column. Hit rate@k: share of cases whose true diagnosis code is among the first k codes of the strategy's differential. nDCG@10 and MRR reward a true diagnosis placed higher. Every case counts in the denominator.



How sure the differences are

Strategy	Hit rate@1 (95% interval)	Hit rate@10 (95% interval)
1. Hypothesis-driven	15.8% (12.0% to 20.6%)	29.4% (24.3% to 35.1%)
2. Knowledge-driven	15.4% (11.6% to 20.2%)	34.2% (28.8% to 40.0%)
3. Channel-based	16.2% (12.3% to 21.0%)	36.4% (30.9% to 42.3%)
4. Sequence-based	19.1% (14.9% to 24.2%)	38.2% (32.7% to 44.1%)

Wilson intervals over the cases of this single run. The strategies ran on the same cases, so the paired comparison below is the sharper test.

Pair (first vs second)	@1: cases only the first / only the second hit	p	@10: only the first / only the second	p
1. Hypothesis-driven vs 2. Knowledge-driven	16 / 15	1.000	18 / 31	0.085
1. Hypothesis-driven vs 3. Channel-based	20 / 21	1.000	16 / 35	0.011
1. Hypothesis-driven vs 4. Sequence-based	19 / 28	0.243	20 / 44	**0.004**
2. Knowledge-driven vs 3. Channel-based	19 / 21	0.875	23 / 29	0.488
2. Knowledge-driven vs 4. Sequence-based	20 / 30	0.203	23 / 34	0.185
3. Channel-based vs 4. Sequence-based	22 / 30	0.332	29 / 34	0.615

Exact McNemar test on the same 272 cases. With 6 comparisons, a p-value below 0.008 (bold) holds after a Bonferroni correction; values between that and 0.05 are suggestive only.

What was run

- 272 cases from `cases/combined_272_whole_chunking.json`, each run with every strategy: 1088 case runs.
- Each strategy could ask at most 4 questions and could stop earlier by its own rule. When it stopped, it gave a differential of up to 10 diagnoses with probabilities.
- Strategy model: anthropic/claude-haiku-5.5. Simulated environment model: openai/gpt-6-luna; literature sources europe_pmc, litsense.
- Coding: ICD-11 MMS 2024-01, first result of the flexible search for the diagnosis name. A prediction counts as a hit only when its code equals the true diagnosis code exactly.
- 11 case runs failed on a token limit in the first pass and were run again (11 of 3. Channel-based); the results use the second attempt.
- Total time 69 min; total cost $5.83 (strategies $2.70, environment $3.14).

How the strategies behaved

Strategy	Questions per case	Why runs stopped	Strategy LLM calls per case	Seconds per case	Strategy cost	Environment cost	Failed runs	No differential
1. Hypothesis-driven	3.94	max iterations 264, strategy done 8	7.9	153	$1.04	$0.77	0	0
2. Knowledge-driven	4.00	max iterations 272	8.0	148	$0.72	$0.77	0	0
3. Channel-based	3.91	max iterations 260, strategy done 12	9.9	152	$0.39	$0.82	0	0
4. Sequence-based	4.00	max iterations 271, strategy done 1	8.0	153	$0.55	$0.78	0	0

Where the answers came from (questions)	Case study	Literature	Unanswerable	No answer (error)
1. Hypothesis-driven	365	657	51	0
2. Knowledge-driven	396	663	29	0
3. Channel-based	309	693	62	0
4. Sequence-based	348	678	61	0

Coding

Strategy	Predicted diagnoses	Without an ICD code	Cases with no coded prediction	Cases whose true diagnosis has no code
1. Hypothesis-driven	2720	5	0	2
2. Knowledge-driven	2720	0	0	2
3. Channel-based	2712	2	0	2
4. Sequence-based	2720	2	0	2

4059 distinct diagnosis names were coded; 7 have no ICD-11 match. A case whose true diagnosis has no code cannot be hit by any strategy.

Reading these numbers

- Exact code match is strict: a correct diagnosis worded so that the search returns a neighbouring code (a parent, a sibling, or a code with a different extension) counts as a miss.
- One run per case and strategy. LLM sampling differs between runs, so small differences between strategies may not repeat.
- The stopping thresholds are starting values and were not tuned.
- Answers marked "literature" are synthesized by the simulated environment when the case text is silent; they are plausible, not facts about the patient.